Type MS · Finding n in a geometric series with logarithms

Answer 1 of 14 · Question 8 part d · 4 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

8 (a)	$a+ar=400$ or $\frac{a(1-r^2)}{1-r}=400$, $ar+ar^2=100$ oe	B1B1
	Common methods 1) $r(a+ar)=100 \Rightarrow r(400)=100$	M1
	$r=\frac{1}{4}$*	A1cso
	2) $a=\frac{400}{1+r} \Rightarrow \left(\frac{400}{1+r}\right)r+\left(\frac{400}{1+r}\right)r^2=100$	M1
	$\Rightarrow 400r+400r^2=100+100r \Rightarrow 4r^2+3r-1=0 \Rightarrow (4r-1)(r+1)(=0)$ $r=\frac{1}{4}$*	A1cso
	3) $r=\frac{400-a}{a} \Rightarrow a\left(\frac{400-a}{a}\right)+a\left(\frac{400-a}{a}\right)^2=100 \Rightarrow$	M1
	$\Rightarrow 400-a+\frac{(400-a)^2}{a}=100 \Rightarrow$ $400-a^2+160000-800a+a^2=100a \Rightarrow a=320 \Rightarrow 320+320r=400$ $r=\frac{1}{4}$*	A1cso [4]
ALT	(Let G_1, G_2, G_3 be the first 3 terms) $G_2+G_3=100$	B1
	$(G_1+G_2=400) \Rightarrow rG_1+rG_2=100$	B1
	$rG_1+rG_2=100 \Rightarrow (r(G_1+G_2)=100) \Rightarrow r(400)=100$	M1
	$r=\frac{1}{4}$ *	A1*cso [4]
(b)	$(a=)\frac{300}{1-\left(\frac{1}{4}\right)^2}$ or $\frac{400}{1+\frac{1}{4}}$ or $\frac{100}{\frac{1}{4}+\left(\frac{1}{4}\right)^2}$ $=320$ *	M1A1 cso [2]
(c)	$S_\infty=\frac{320}{1-\frac{1}{4}}=\frac{1280}{3}$	M1A1 [2]
		Total 12 marks

Type MS · Finding n in a geometric series with logarithms

Answer 1 of 14 · Question 8 part d · 4 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part d in this mark scheme extract

Part	Mark	Notes
(a)	B1	For either equation shown correct a and r can be any letters throughout.
	B1	For both equations shown correct a and r can be any letters throughout.
	M1	For forming an equation eliminating a or r Allow one error in processing **such as a sign or arithmetical error, but not a 'cancellation'/simplification error**. Must be working with 2 correct equations. This mark can be awarded as soon as a or r are eliminated. Doesn't need simplification at this stage.
	A1 cso	For correctly solving and attaining $r = \frac{1}{4}$ minimum steps shown, no errors/omissions, ignore $r = -1$
ALT	B1	For either equation shown correct
	B1	For both equations shown correct
	M1	For multiplication of the first equation by r and formation of an equation in r Allow one error in processing. Must be working with 2 correct equations.
	A1	For $r = \frac{1}{4}$ minimum steps shown, no errors or omissions. Ignore work on any negative values
There are a number of different methods to do this, the four most commonly anticipated are shown. Mark to the following principles to gain the method mark: • One processing error only in any method (**M mark only, not A mark).** • Rearrange for r or a and correctly substitute into the other equation or such as method 1 to reach an equation in one variable only. • Rearrange the resulting equation so that an equation of the form $\boldsymbol{br = c}$ is reached. Note for the quadratic option, a factorisation will suffice. Note, if eliminating r, a must be found and the value of a then substituted into an appropriate equation. Methods where these principles can't be applied and thought worthy of credit – **send to review please**.		
(b)	M1	For using their expression for a with the correct r, to find a value for a Note, **for this question only**, work in (a) may be credited for this mark – **only if they eliminated r in their solution for part (a) and this is then used.**
	A1cso	For 320, no errors.
(c)	M1	For using the correct formula for the sum to infinity of a convergent series with the given values of a and r to find a value.
	A1	For the exact value of $\frac{1280}{3}$ oe or 426.67 or better (ie correctly rounded to more decimal places) or 426.6… (minimum 3 dots) or 426.66^{r} or $426.\dot{6}$
(d)	M1	Uses the correct formula for the sum of a geometric series, to set up **an inequality or equation,** allow < or > or = using the given values of r and a. Condone $\frac{1}{4}^{n}$
	dM1	For simplifying (allow errors in simplification) their **inequality or equation** in n to the form $\left(\frac{1}{4}\right)^{n} < d \quad d \neq 0$ or $4^{n} < d$ Allow < or > or =. Dependent on the 1st method mark. Condone poor bracketing with powers again.
	ddM1	For the correct use of logs and correct use of an inequality sign throughout, including the reversal at the appropriate point. Dependent on both previous method marks. This mark may not be awarded if 'd' is negative. If candidates give a final answer of (n =) 7 – this mark can be implied even if the inequality sign is not correctly reversed.
	A1	For (n =) 7 Note although $n = 7$ can imply ddM1 as described, it is unlikely to imply the first 2 marks as there must be some logs work (directed by the question).

Type MS · Finding n in a geometric series with logarithms

Answer 1 of 14 · Question 8 part d · 4 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part d in this mark scheme extract

(d)	$426.6 < \dfrac{320\left(1-\left(\frac{1}{4}\right)^n\right)}{1-\frac{1}{4}} \Rightarrow \left(\frac{1}{4}\right)^n < \frac{1}{6400}$ or $4^n < 6400$	M1dM1
	$\Rightarrow n > \dfrac{\log\left(\frac{1}{6400}\right)}{\log\left(\frac{1}{4}\right)}$ or $n > \log_{\frac{1}{4}}\left(\frac{1}{6400}\right)$ oe	ddM1
	$\Rightarrow n > 6.32... \Rightarrow n = 7$	A1 [4]
		Total 12 marks

Type MS · Finding n in a geometric series with logarithms

Answer 1 of 14 · Question 8 part d · 4 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding n in a geometric series with logarithms

Answer 2 of 14 · Question 7 part b · 4 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

7 (a)	(i) $\frac{U_7}{U_3}=\frac{ar^6}{ar^2}=r^4\Rightarrow\frac{16}{4}=r^4\Rightarrow 4=r^4\Rightarrow 2=r^2\Rightarrow r=\sqrt{2}*$ (ii) $ar^2=4\Rightarrow a=\frac{4}{\left(\sqrt{2}\right)^2}=2$	M1 A1cso A1 [3]
Alternative		
(a)	(i) $ar^2=4\ ,\ ar^6=16\Rightarrow a\left(\frac{4}{a}\right)^3=16\Rightarrow a^2=4\Rightarrow a=2$ $r^2=\frac{4}{2}\Rightarrow r=\sqrt{2}*$ (ii) $a=2$	M1 A1cso A1
(b)	$500<2\times\sqrt{2}^{(n-1)}\Rightarrow 250<\sqrt{2}^{(n-1)}$ or $500<2^{\frac{1}{2}(n+1)}$ $\lg 250<(n-1)\lg\sqrt{2}$ or $\lg 500<\frac{1}{2}(n+1)\lg 2$ $n-1>\frac{\lg 250}{\lg\sqrt{2}}\Rightarrow n-1>15.93...$ or $\frac{1}{2}(n+1)>\frac{\lg 500}{\lg 2}\Rightarrow\frac{1}{2}(n+1)>8.96...$ $n>16.93...\Rightarrow n=17$	M1 M1 M1A1 [4]
(c)	$S_{20}=\frac{2\left(\left(\sqrt{2}\right)^{20}-1\right)}{\sqrt{2}-1}=\frac{2046}{\sqrt{2}-1}$ $\frac{2046}{\left(\sqrt{2}-1\right)}\times\frac{\left(\sqrt{2}+1\right)}{\left(\sqrt{2}+1\right)}=2046\sqrt{2}+2046=2046\left(1+\sqrt{2}\right)$	M1A1 M1A1 [4]
		Total 11 marks

(a)(i)	M1	$ar^2=4$ and $ar^6=16$
	A1 cso	Correct working to eliminate a and show that $r=\sqrt{2}$ (Answer given) Must see $r^4=4$.
(ii)	A1	$a=2$ (NB: This is not a B mark; must have scored the first M mark.)
Alternative		
(a)(i)	M1	$ar^2=4$ and $ar^6=16$
	A1 cso	Correct working to find a and use it to show that $r=\sqrt{2}$ (Answer given) Must see a correct substitution to eliminate r which is processed correctly to give $a=2$, and $r^2=\frac{4}{2}$ or $r^6=\frac{16}{2}$.
(ii)	A1	$a=2$ (Must have scored the first M mark.)

Type MS · Finding n in a geometric series with logarithms

Answer 2 of 14 · Question 7 part b · 4 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

(b)	M1	$500 < 2\times\left(\sqrt{2}\right)^{(n-1)}$ Accept an equation or >. Accept $500 < 2\times\sqrt{2}^{n-1}$ ft their value for a
	M1	$n-1 > \frac{\lg 250}{\lg\sqrt{2}}$ or $\frac{1}{2}(n-1) > \frac{\lg 250}{\lg 2}$ or $\frac{1}{2}(n+1) > \frac{\lg 500}{\lg 2}$ or $\lg_{\sqrt{2}} 250 < n-1$ Accept equation or >, and logs with any base. ft $500 < 2\times\left(\sqrt{2}\right)^{n}$
	M1	Evaluate and divide logs and attempt to find a value (at least 1DP) or numerical expression for n. Ignore incorrect inequality sign. eg $n > 16.9$ or $n = 16.9$ or $n > 1+15.93$ or $n = 1+2\times7.96$ ft $500 < 2\times\left(\sqrt{2}\right)^{n}$
	A1	$n = 17$ Do not award after incorrect inequalities e.g. $n < 16.9$
Alternative		
(b)	M1	$500 < 2\times\left(\sqrt{2}\right)^{(n-1)}$ Accept an equation or >. Accept $500 < 2\times\sqrt{2}^{n-1}$ ft their value for a
	M1	Attempt to use trial and improvement, showing two trials that can be used to confirm the required value of n, eg $\left(\sqrt{2}\right)^{15} = 181.0...$ and $\left(\sqrt{2}\right)^{16} = 256$ or $2\left(\sqrt{2}\right)^{15} = 362.0...$ and $2\left(\sqrt{2}\right)^{16} = 512$ or $2^8 = 256$ and $2^9 = 512$ ft $500 < 2\times\left(\sqrt{2}\right)^{n}$ if clearly stated
	M1	Use the higher power from these trials in an appropriate equation eg $n-1 = 16$ or $\frac{1}{2}(n+1) = 9$ and attempt to find a value for n. ft $500 < 2\times\left(\sqrt{2}\right)^{n}$ if clearly stated
	A1	$n = 17$
(c)	M1	Attempt to use $S_n = \frac{a(1-r^n)}{(1-r)}$ with their a, $r = \sqrt{2}$ and $n = 20$ Allow notation or sign errors if formula is stated, otherwise the substitution must be correct.
	A1	$S_{20} = \frac{2046}{\sqrt{2}-1}$ or $S_{20} = -\frac{2046}{1-\sqrt{2}}$ Correct expression with numerator evaluated.
	M1	Attempt to rationalise the denominator. Must use $\sqrt{2}+1$ and attempt to simplify the denominator.
	A1	$2046(1+\sqrt{2})$ or $2046(\sqrt{2}+1)$ Accept $p = 2046$. Award full marks for a correct answer if first M1 has been awarded.

Type MS · Finding n in a geometric series with logarithms

Answer 2 of 14 · Question 7 part b · 4 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding n in a geometric series with logarithms

Answer 3 of 14 · Question 9 part b · 3 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

9(a)	$ar^2 = \frac{27}{2}$	B1
	$a + ar + \frac{27}{2} = \frac{57}{2}\left(\Rightarrow a + ar = 15 \Rightarrow a(1+r) = 15 \Rightarrow a = \frac{15}{1+r}\right)$	M1
	$\left(\frac{15}{1+r}\right)r^2 = \frac{27}{2} \Rightarrow 30r^2 - 27r - 27 = 0 \Rightarrow (10r^2 - 9r - 9 = 0)$	M1
	ALT $ar^2 = \frac{27}{2}$	B1
	$\frac{57}{2} = \frac{a(r^3-1)}{r-1}$	M1
	$\frac{57}{2} = \frac{27}{2r^2} \times \frac{(r^3-1)}{r-1} \Rightarrow 30r^3 - 57r^2 + 27 = 0$ $\left(\Rightarrow 10r^3 - 19r^2 + 9 = 0\right)$	M1
	r: $10r^2 - 9r - 9 = 0 \Rightarrow (5r+3)(2r-3) = 0 \Rightarrow r = \frac{3}{2}$	M1A1
	a: $a = \frac{15}{1+\frac{3}{2}} = 6$	A1
	$S_n = \sum_{r=1}^{n}\left('6' \div \frac{3}{2}\right)\left(\frac{3}{2}\right)^r \Rightarrow S_n = \sum_{r=1}^{n} 4\left(\frac{3}{2}\right)^r$ *	M1A1 cso [8]
(b)	$\frac{6\left(1.5^k - 1\right)}{1.5-1} > 50\,000$	M1
	$1.5^k > \frac{12\,503}{3} \Rightarrow \lg 1.5^k > \lg\frac{12\,503}{3}$	M1
	$k > \frac{\lg\frac{12\,503}{3}}{\lg\frac{3}{2}}$ or $k > \frac{\lg\frac{12\,503}{3}}{\lg 1.5}$ *	A1 [cso] [3]

Type MS · Finding n in a geometric series with logarithms

Answer 3 of 14 · Question 9 part b · 3 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

(c)	For $k > 20.556...$ so $k = 21$	B1 [1]
		Total 12 marks

Question	Scheme	Marks
9(a)	For stating, $ar^2 = \frac{27}{2}$	B1
	For summing the first three terms; $a + ar + \frac{27}{2} = \frac{57}{2}\left(\Rightarrow a + ar = 15 \Rightarrow a(1+r) = 15 \Rightarrow a = \frac{15}{1+r}\right)$ OR $\frac{57}{2} = \frac{a(r^3-1)}{r-1}$	M1
	For attempting to form a 3TQ using their two expressions for U_3 and S_3 $\left(\frac{15}{1+r}\right)r^2 = \frac{27}{2} \Rightarrow 30r^2 - 27r - 27 = 0 \Rightarrow (10r^2 - 9r - 9 = 0)$ OR For attempting to form a cubic using their two expressions for U_3 and S_3 $\frac{57}{2} = \frac{27}{2r^2} \times \frac{(r^3-1)}{r-1} \Rightarrow 30r^3 - 57r^2 + 27 = 0$	M1
	For attempting to solve their 3TQ (see general guidance). $10r^2 - 9r - 9 = 0 \Rightarrow (5r+3)(2r-3) = 0 \Rightarrow r = ...$ OR If following the alt method then this mark is awarded for factorising the cubic into a linear expression and a 3TQ **and** attempting to solve their 3TQ. $(r-1)(10r^2 - 9r - 9) = 0 \Rightarrow (r-1)(5r+3)(2r-3) = 0$ $\Rightarrow r = \cdots$	M1
	For the correct value of $r = \frac{3}{2}$	A1
	For the correct value of $a = \frac{15}{1+\frac{3}{2}} = 6$	A1
	For attempting to use their a and r to find the correct expression for S_n	

Type MS · Finding n in a geometric series with logarithms

Answer 3 of 14 · Question 9 part b · 3 marks

2023 JAN · Paper 02 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

	$S_n = \sum_{r=1}^{n}\left('6' \div \frac{3}{2}\right)\left(\frac{3}{2}\right)^r \Rightarrow S_n = \ldots$ OR For demonstrating that a particular value of r gives the correct term in the sequence.	M1
	For the correct expression $S_n = \sum_{r=1}^{n} 4\left(\frac{3}{2}\right)^r$ * OR For demonstrating that a particular value of r gives the correct term in the sequence and commenting on a correct common ratio.	A1 cso [8]
(b)	For using the summation formula > 50 000 $\frac{6\left(1.5^k - 1\right)}{1.5 - 1} > 50\ 000$	M1
	For rearranging the inequality to achieve, $1.5^k > \frac{12\ 503}{3}$ and takes logs base 10 of both sides $\lg 1.5^k > \lg \frac{12\ 503}{3}$	M1
	For using the laws of logs to make k the subject. $k > \frac{\lg \frac{12\ 503}{3}}{\lg \frac{3}{2}}$ or $k > \frac{\lg \frac{12\ 503}{3}}{\lg 1.5}$ *	A1 cso [3]
(c)	For $k > 20.556\ldots \Rightarrow k = 21$	B1 [1]
		Total 12 marks

Type MS · Finding n in a geometric series with logarithms

Answer 3 of 14 · Question 9 part b · 3 marks

2023 JAN · Paper 02 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding n in a geometric series with logarithms

Answer 4 of 14 · Question 9 part c · 1 mark

2023 JAN · Paper 02 (regional R) · extract 1 of 4 · find part c in this mark scheme extract

9(a)	$ar^2 = \frac{27}{2}$ $a + ar + \frac{27}{2} = \frac{57}{2}\left(\Rightarrow a + ar = 15 \Rightarrow a(1+r) = 15 \Rightarrow a = \frac{15}{1+r}\right)$ $\left(\frac{15}{1+r}\right)r^2 = \frac{27}{2} \Rightarrow 30r^2 - 27r - 27 = 0 \Rightarrow (10r^2 - 9r - 9 = 0)$	B1 M1 M1
	ALT $ar^2 = \frac{27}{2}$ $\frac{57}{2} = \frac{a(r^3-1)}{r-1}$ $\frac{57}{2} = \frac{27}{2r^2} \times \frac{(r^3-1)}{r-1} \Rightarrow 30r^3 - 57r^2 + 27 = 0$ $\left(\Rightarrow 10r^3 - 19r^2 + 9 = 0\right)$	B1 M1 M1
	r: $10r^2 - 9r - 9 = 0 \Rightarrow (5r+3)(2r-3) = 0 \Rightarrow r = \frac{3}{2}$ a: $a = \frac{15}{1+\frac{3}{2}} = 6$ $S_n = \sum_{r=1}^{n}\left('6' \div \frac{3}{2}\right)\left(\frac{3}{2}\right)^r \Rightarrow S_n = \sum_{r=1}^{n} 4\left(\frac{3}{2}\right)^r$ *	M1A1 A1 M1A1 cso [8]
(b)	$\frac{6\left(1.5^k - 1\right)}{1.5-1} > 50\,000$ $1.5^k > \frac{12\,503}{3} \Rightarrow \lg 1.5^k > \lg\frac{12\,503}{3}$ $k > \frac{\lg\frac{12\,503}{3}}{\lg\frac{3}{2}}$ or $k > \frac{\lg\frac{12\,503}{3}}{\lg 1.5}$ *	M1 M1 A1 [cso] [3]

Type MS · Finding n in a geometric series with logarithms

Answer 4 of 14 · Question 9 part c · 1 mark

2023 JAN · Paper 02 (regional R) · extract 2 of 4 · find part c in this mark scheme extract

(c)	For $k > 20.556...$ so $k = 21$	B1 [1]
		Total 12 marks

Question	Scheme	Marks
9(a)	For stating, $ar^2 = \frac{27}{2}$	B1
	For summing the first three terms; $a + ar + \frac{27}{2} = \frac{57}{2}\left(\Rightarrow a + ar = 15 \Rightarrow a(1+r) = 15 \Rightarrow a = \frac{15}{1+r}\right)$ OR $\frac{57}{2} = \frac{a(r^3-1)}{r-1}$	M1
	For attempting to form a 3TQ using their two expressions for U_3 and S_3 $\left(\frac{15}{1+r}\right)r^2 = \frac{27}{2} \Rightarrow 30r^2 - 27r - 27 = 0 \Rightarrow (10r^2 - 9r - 9 = 0)$ OR For attempting to form a cubic using their two expressions for U_3 and S_3 $\frac{57}{2} = \frac{27}{2r^2} \times \frac{(r^3-1)}{r-1} \Rightarrow 30r^3 - 57r^2 + 27 = 0$	M1
	For attempting to solve their 3TQ (see general guidance). $10r^2 - 9r - 9 = 0 \Rightarrow (5r+3)(2r-3) = 0 \Rightarrow r = ...$ OR If following the alt method then this mark is awarded for factorising the cubic into a linear expression and a 3TQ **and** attempting to solve their 3TQ. $(r-1)(10r^2 - 9r - 9) = 0 \Rightarrow (r-1)(5r+3)(2r-3) = 0$ $\Rightarrow r = \cdots$	M1
	For the correct value of $r = \frac{3}{2}$	A1
	For the correct value of $a = \frac{15}{1+\frac{3}{2}} = 6$	A1
	For attempting to use their a and r to find the correct expression for S_n	

Type MS · Finding n in a geometric series with logarithms

Answer 4 of 14 · Question 9 part c · 1 mark

2023 JAN · Paper 02 (regional R) · extract 3 of 4 · find part c in this mark scheme extract

	$S_n = \sum_{r=1}^{n}\left('6' \div \frac{3}{2}\right)\left(\frac{3}{2}\right)^r \Rightarrow S_n = \ldots$ OR For demonstrating that a particular value of r gives the correct term in the sequence.	M1
	For the correct expression $S_n = \sum_{r=1}^{n} 4\left(\frac{3}{2}\right)^r$ * OR For demonstrating that a particular value of r gives the correct term in the sequence and commenting on a correct common ratio.	A1 cso [8]
(b)	For using the summation formula > 50 000 $\frac{6\left(1.5^k - 1\right)}{1.5 - 1} > 50\ 000$	M1
	For rearranging the inequality to achieve, $1.5^k > \frac{12\ 503}{3}$ and takes logs base 10 of both sides $\lg 1.5^k > \lg \frac{12\ 503}{3}$	M1
	For using the laws of logs to make k the subject. $k > \frac{\lg \frac{12\ 503}{3}}{\lg \frac{3}{2}}$ or $k > \frac{\lg \frac{12\ 503}{3}}{\lg 1.5}$ *	A1 cso [3]
(c)	For $k > 20.556\ldots \Rightarrow k = 21$	B1 [1]
		Total 12 marks

Type MS · Finding n in a geometric series with logarithms

Answer 4 of 14 · Question 9 part c · 1 mark

2023 JAN · Paper 02 (regional R) · extract 4 of 4 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding n in a geometric series with logarithms

Answer 5 of 14 · Question 7 part d · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

7(a)	$\frac{(x+1)}{(x-3)}=\frac{(4x-2)}{(x+1)}$ or $(x+1)^2=(x-3)(4x-2)$	M1
	$(x+1)^2=(4x-2)(x-3)\Rightarrow 3x^2-16x+5(=0)$	A1
	$(x-5)(3x-1)=0\Rightarrow x=5,\ \frac{1}{3}$	M1A1A1 (5)
(b)	$x=\frac{1}{3}\Rightarrow r=\frac{x+1}{x-3}=\frac{\frac{4}{3}}{\frac{-8}{3}}$	M1
	$r=-\frac{1}{2}$ ∴ convergent as $-1<r<1$ *	A1cso (2)
(c)	$S=\frac{a}{1-r}=\frac{-\frac{8}{3}}{1+\frac{1}{2}}=-\frac{16}{9}$ oe	M1A1 (2)
(d)	$\frac{S}{S_n}=\frac{a}{1-r}\times\frac{1-r}{a(1-r^n)}=\frac{1}{(1-r^n)}=\frac{256}{255}$	M1
	$255=256(1-r^n)$	
	$256r^n=1$ $\left(-\frac{1}{2}\right)^n=\frac{1}{256}$ oe $n=8$	dM1A1 (3) [12]

(a)M1	Form an equation using the given information about the terms
A1	Simplify their equation to a correct 3TQ, terms in any order Condone missing = 0
M1	Attempt to solve their 3TQ by any valid method. Must reach x = ... (at least one root) Calculator solutions: *Both* roots correct from a correct equation scores M1A1A1 Incorrect equation or incorrect roots scores M0A0A0
A1	One correct value of x
A1	Both correct values of x
(b)	
M1	Use either of their values of x, provided it is < 1, to find the corresponding value of r. No need to simplify
A1cso	Correct value of r **and** the conclusion including the reason
(c)	
M1	Use their value of r (not x) provided $-1<r<1$ (as found in (b) or here) and the formula for the sum to infinity to obtain a value for S
A1	Correct value
(d)	
M1	Obtain an equation in r and n. May use the formulae to cancel a or may sub values of a and r in the formulae for the LHS Must equate to 256/255 Value of r not needed for this mark so allow any value used.
dM1	Solve their equation of the form $r^n=...$ where $-1<r<1$ (r not x). May use trial and improvement or logs. This mark can be given if the equation and value of r are incorrect. Evidence of method needed if final answer is incorrect. If logs used condone log $(-1/2)^n$
A1	Correct value from correct working

Type MS · Finding n in a geometric series with logarithms

Answer 5 of 14 · Question 7 part d · 3 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question Number	Scheme	Marks
	5	

Type MS · Finding n in a geometric series with logarithms

Answer 6 of 14 · Question 8 part c · 6 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

8(a)	$U_5 = \left(\frac{25}{4}\right)\left(\frac{3}{5}\right)^5 = \frac{243}{500}$ oe	B1 [1]
(b)	$U_1 = \left(\frac{25}{4}\right)\left(\frac{3}{5}\right)^1 = \frac{15}{4}$ oe	B1
	$S_n = \left(\frac{15}{4}\right)\left(\frac{3}{5}\right)^0 + \left(\frac{15}{4}\right)\left(\frac{3}{5}\right)^1 + \ldots\ldots + \left(\frac{15}{4}\right)\left(\frac{3}{5}\right)^{n-1}$	M1
	$\Rightarrow S_n = \sum_{r=1}^{n}\left(\frac{15}{4}\right)\left(\frac{3}{5}\right)^{r-1}$ oe	A1 [3]
	ALT 1	
	$U_1 = \left(\frac{25}{4}\right)\left(\frac{3}{5}\right)^1 = \frac{15}{4}$	[B1
	$r = \frac{3}{5} \qquad U_n = \frac{15}{4} \times \left(\frac{3}{5}\right)^{n-1}$	M1
	$S_n = \sum_{r=1}^{n} \frac{15}{4} \times \left(\frac{3}{5}\right)^{r-1}$	A1]
	ALT 2	
	$S_n = \sum_{r=1}^{n} \frac{25}{4}\left(\frac{3}{5}\right)^r$	[B1
	$= \sum_{r=1}^{n} \frac{25}{4} \times \frac{3}{5} \times \left(\frac{3}{5}\right)^{r-1}$	M1
	$= \sum_{r=1}^{n} \frac{15}{4} \times \left(\frac{3}{5}\right)^{r-1}$	A1]
(c)	$S_\infty = \frac{\frac{15}{4}}{1-\frac{3}{5}} = \frac{75}{8}$ or $9.375 \qquad S_n = \frac{\frac{15}{4}\left(1-\left[\frac{3}{5}\right]^n\right)}{1-\frac{3}{5}}$	M1M1
	$9.375 - \frac{\frac{15}{4}\left(1-\left[\frac{3}{5}\right]^n\right)}{1-\frac{3}{5}} < 0.045 \Rightarrow \frac{\frac{15}{4}\left(1-\left[\frac{3}{5}\right]^n\right)}{1-\frac{3}{5}} > 9.33$	
	$\frac{\frac{15}{4}\left(1-\left[\frac{3}{5}\right]^n\right)}{1-\frac{3}{5}} > 9.33 \Rightarrow 1-\left[\frac{3}{5}\right]^n > 0.9952 \Rightarrow \left[\frac{3}{5}\right]^n < \frac{3}{625}$	dM1
	$n \lg\left[\frac{3}{5}\right] < \lg\frac{3}{625} \Rightarrow n > 10.45$	ddM1A1
	$n = 11$	A1 [6]
		Total 10 marks

Type MS · Finding n in a geometric series with logarithms

Answer 6 of 14 · Question 8 part c · 6 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
		Mark (a) and (b) together
(a)	B1	For the correct value only. Decimal value is 0.486.
(b)	B1	Finds the value of the first term. Must be identified as U_1 / first term.
	M1	Forms the sum of the series with at least the first term and the nth term.
	A1	For the correct expression.
	ALT 1	
	B1	Finds the value of the first term. Must be identified as U_1 / first term.
	M1	For finding r and writing $U_n = \frac{15}{4} \times \left(\frac{3}{5}\right)^{n-1}$
	A1	For the correct expression.
	ALT 2	
	B1	For correct summation statement using U_n from the question
	M1	For showing process to change index from r to $r-1$ and find the value of $\frac{A}{B}$ Minimum working is $\frac{25}{4} \times \left(\frac{3}{5}\right)^{r} = \frac{25}{4} \times \frac{3}{5} \times \left(\frac{3}{5}\right)^{r-1}$. May be with n as index.
	A1	For the correct expression.
(c)	Trial and improvement approaches for (c) should be sent to review.	
	M1	Correct method to find the sum to infinity using the correct formula with their $a = \frac{15}{4}$ and $r = \frac{3}{5}$. Their a should be $\frac{A}{B}$ if found in (b). Condone if working with $a = \frac{25}{4}$, r must be correct.
	M1	Using the correct formula, forms the sum to n terms using their $a = \frac{15}{4}$ and $r = \frac{3}{5}$. Their a should be $\frac{A}{B}$ if found in (b). Condone if working with $a = \frac{25}{4}$, r must be correct.
	dM1	Forms the inequality in terms of n with their S_∞ and S_n and simplifies to $\left[\frac{3}{5}\right]^n < \frac{3}{625}$ Must have dealt with negative. Allow for working with equation if inequality sign reinstated later. Reinstatement can be implied by rounding up to the next integer value once n found. Dep M1M1
	ddM1	Uses logarithms correctly to attempt to find the value of n They must reverse the inequality for this mark. Allow for working with equation if inequality sign reinstated later. Reinstatement can be implied by rounding up to the next integer value once n found. Dep previous M mark. Allow use of log base $\frac{3}{5}$.
	A1	For correct n awrt 10.5 Can be implied by $n = 11$ if decimal not seen.
	A1	For $n = 11$

Type MS · Finding n in a geometric series with logarithms

Answer 6 of 14 · Question 8 part c · 6 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding n in a geometric series with logarithms

Answer 7 of 14 · Question 8 part d · 4 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part d in this mark scheme extract

8(a)	$a + ar = 360$ oe e.g. $\frac{a(1-r^2)}{1-r} = 360$ $ar + ar^2 = 288$ oe $r = \left\{\frac{a(1+r)}{ar(1+r)} =\right\} \frac{288}{360} = \frac{4}{5}$ $a = \frac{360}{1 + "\frac{4}{5}"} = 200$ $U_n = 200\left(\frac{4}{5}\right)^{n-1}$ *	M1M1 (B1B1on ePen dM1A1 M1A1 A1cso(B1 on ePen) [7]
8(a) ALT	$a + ar = 360$ oe e.g. $\frac{a(1-r^2)}{1-r} = 360$ $ar + ar^2 = 288$ oe $a\left(\frac{360}{a} - 1\right) + a\left(\frac{360}{a} - 1\right)^2 = 288$ $\Rightarrow a = 200$ $r = \frac{360}{"200"} - 1 = \frac{4}{5}$ $U_n = 200\left(\frac{4}{5}\right)^{n-1}$ *	M1M1 (B1B1 on ePen) dM1A1 M1A1 A1cso(B1 on ePen) [7]
(b)	{S is convergent because} $\left\|\frac{4}{5}\right\| < 1$	B1 [1]
(c)	$S_\infty = \frac{'200'}{1-'\left(\frac{4}{5}\right)'} = 1000$	M1A1cao [2]
(d)	$\frac{200\left(1-\left(\frac{4}{5}\right)^n\right)}{1-\left(\frac{4}{5}\right)} > 978$ $\left(\frac{4}{5}\right)^n < \frac{11}{500}$ $n\,log\left(\frac{4}{5}\right) < log\left(\frac{11}{500}\right)$ $\Rightarrow n > 17.104\ldots$ $n = 18$	M1 dM1 ddM1 A1 [4]
		Total 14 marks

Type MS · Finding n in a geometric series with logarithms

Answer 7 of 14 · Question 8 part d · 4 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part d in this mark scheme extract

Part	Mark	Notes
(a)	M1	For either correct equation. Allow use of U_1 as the first term instead of a
	M1	For both correct equations. Allow use of U_1 as the first term instead of a
	dM1	For eliminating a from their equations, reaches r =... (Depends on previous mark)
	A1	For the correct value of r
	M1	Uses their value of r to find a value for a
	A1	For a = 200
	A1 cso	For the correct required expression of $\boldsymbol{U_n = 200\left(\frac{4}{5}\right)^{n-1}}$ (must have $\boldsymbol{U_n}$)
(a)ALT	M1	For either correct equation.
	M1	For both correct equations
	dM1	For eliminating r from their equations, reaches a =... (Depends on previous mark)
	A1	For the correct value of a
	M1	Uses their value of a to find a value for r
	A1	For the correct value of r
	A1 cso	For the correct required expression of $U_n = 200\left(\frac{4}{5}\right)^{n-1}$ (must have $\boldsymbol{U_n}$)
(b)	B1	For stating the correct reason, $\lvert r\rvert < 1$ or -1 < r < 1 or "0.8" < 1 or $\left\lvert "\frac{4}{5}" \right\rvert < 1$ **Do not accept** just r < 1 without a correct reason
(c)	M1	For using their value of A /a /first term of their geometric sequence and their r (provided their $\lvert r\rvert < 1$)in a correct formula for the sum to infinity. $S_\infty = \frac{'their\ A'}{1-'(their\ r)'}$
	A1cao	For the correct value of 1000
(d)	M1	Uses their a and r (their r can be $\lvert r\rvert < 1$ or $\lvert r\rvert > 1$) to set the correct formula for the sum of a geometric series > 978 or =978 or ≥ 978 (>978 or =978 or ≥ 978 may be implied by later work.)
	dM1	Attempts to rearrange and achieves $(their\ r)^n$ " < "k (k *is non zero positive constant*) **Allow strict or non-strict inequalities or equation**: <,>, ≥ , ≤ or =
	ddM1	For correctly takes logs (any base)/ln, and achieves a positive value for n, it maybe implied by 17.1…but not 18 **ALT**: trial and error to find a value of n (at least tries n=17 and n=18)
	A1	For n = 18 Must come from **correct** working, achieves n=18 with no incorrect inequalities in their working.

If a candidate does not score the dM1 mark but arrives at an answer of 17.1...
this scores a maximum of M1 as clear use of calculator is not condoned.

Type MS · Finding n in a geometric series with logarithms

Answer 7 of 14 · Question 8 part d · 4 marks

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding n in a geometric series with logarithms

Answer 8 of 14 · Question 6 part b.ii · 5 marks

2025 NOV · Paper 02 · extract 1 of 3 · find part b.ii in this mark scheme extract

6(i)(a)	$\left[\frac{ar}{a} = r = \right] \frac{\sqrt{3}}{3+\sqrt{3}} \times \frac{3-\sqrt{3}}{3-\sqrt{3}}$ oe or $\frac{\sqrt{3}\left(3-\sqrt{3}\right)}{3^2-\left(\sqrt{3}\right)^2}$ $= \frac{3\sqrt{3}-3}{6}$ oe $\Rightarrow \frac{\sqrt{3}-1}{2}$ or $\frac{-1+\sqrt{3}}{2}$	M1 A1 [2]
(b) (i)	$\frac{\sqrt{3}-1}{2}\left(=0.366\right)<1$ or $\lvert r\rvert<1$ so the series is convergent	B1ft [1]
(ii)	$S = \frac{8\left(1-0.6^n\right)}{1-0.6}$, $T = \frac{8}{1-0.6} = [20]$ $\left["20"-"20"\left(1-0.6^n\right)<0.12\right] \Rightarrow 0.6^n < 0.006$ oe eg $-0.6^n > -\frac{3}{500}$ $n\log_{10} 0.6 < \log_{10} 0.006$ oe eg $n\log_{0.006} 0.6 < \log_{0.006} 0.006$ $n > \frac{\log_{10} 0.006}{\log_{10} 0.6}$ $\left(n > 10.015\right)$ $n = 11$	M1, M1 M1 dM1 A1 [5]
		Total 8 marks

Type MS · Finding n in a geometric series with logarithms

Answer 8 of 14 · Question 6 part b.ii · 5 marks

2025 NOV · Paper 02 · extract 2 of 3 · find part b.ii in this mark scheme extract

Part	Mark	Notes
(a)	M1	For dividing the second term by the first term and for a correct method **stated** to rationalise the denominator. A correct answer may not imply this mark as candidates are told to show working.
	A1	For $\frac{3\sqrt{3}-3}{6}$ oe **and** $\frac{\sqrt{3}-1}{2}$ or $\frac{-1+\sqrt{3}}{2}$ Accept $p = -1$ and $q = 2$ Minimum steps as shown in the mark scheme needed, with no errors or omissions. Additional steps shown must be checked and correct.
If students start completing 1st term divide 2nd ie $\frac{3+\sqrt{3}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{3\sqrt{3}+3}{3}$ they must progress to $\frac{3}{3\sqrt{3}+3}$ oe eg $\frac{1}{\sqrt{3}+1}$ before marks can be awarded.		
(b) (i)	B1ft	For a correct explanation referring to r or the value being < 1 or $\lvert r \rvert < 1$ This may ft any value of r from their part (a) where $-1 < r < 1$ Accept "r is between 1 and – 1"
(ii)	M1	For the correct expression for S or T using the given values. **May be implied/embedded within the solution.**
	M1	For the correct expression for S and T using the given values. **May be implied/embedded within the solution.**
	M1	For an inequality of the form $0.6^n < A \quad A > 0$ or $-0.6^n > -B \quad B > 0$ $0.6^n < 0.006$ oe eg $-0.6^n > -\frac{3}{500}$ will definitely imply first 2 M marks if one or both are not explicitly stated.
	dM1	For using logarithms correctly, including reversing the inequality to find a value for n Dependent on previous method mark
	A1	For the correct value of n **following correct work.** Candidates must use logarithms and we must see all previous method marks. If ($n = 11$) comes from incorrect logs work or incorrect reversal of this inequality, this will be A0.

Type MS · Finding n in a geometric series with logarithms

Answer 8 of 14 · Question 6 part b.ii · 5 marks

2025 NOV · Paper 02 · extract 3 of 3 · find part b.ii in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding n in a geometric series with logarithms

Answer 9 of 14 · Question 7 part d · 4 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 4 · find part d in this mark scheme extract

7 (a)	$\frac{ar^6}{ar^3} = r^3 = \frac{e^{\frac{2x+1}{2}}}{e^{x+2}} = \frac{e^x \times e^{\frac{1}{2}}}{e^x \times e^2} = e^{-\frac{3}{2}} \Rightarrow r = e^{-\frac{1}{2}}$ * **ALT** $\frac{e^{x+2}}{r^3} r^6 = e^{\frac{2x+1}{2}} \Rightarrow r^3 = e^{x+\frac{1}{2}-x+2} \Rightarrow r^3 = e^{-\frac{3}{2}} \Rightarrow r = e^{-\frac{1}{2}}$ *	M1M1A1 cso [3] {M1} {M1}{A1} cso [3]
(b)	$ar^3 = ae^{-\frac{3}{2}} = e^{x+2} \Rightarrow a = \frac{e^{x+2}}{e^{-\frac{3}{2}}} = \frac{e^x \times e^2}{e^{-\frac{3}{2}}} = e^{x+\frac{7}{2}}$ oe **ALT** $a = \frac{e^{x+2}}{e^{-\frac{3}{2}}} = e^{x+2+\frac{3}{2}} = e^{x+\frac{7}{2}}$ oe	M1M1A1 [3] {M1}{M1} {A1} [3]

(a)	
M1	Use of $\frac{ar^6}{ar^3} = r^3$
M1	Using $e^{a+b} = e^a \times e^b$ to simplify leading to $r^3 = e^c$ where c is a number
A1 cso	Obtains the **given** answer with no errors in the working
ALT	
M1	For rearranging to make a subject and substituting into the other equation
M1	Using $e^{a-b} = e^a \div e^b$ to simplify leading to $r^3 = e^c$ where c is a number
A1 cso	Obtains the **given** answer with no errors in the working
(b)	
M1	For $a = \frac{e^{x+2}}{e^{-\frac{3}{2}}}$
M1	Using $e^{a+b} = e^a \times e^b$ to simplify to $a = \frac{e^x \times e^2}{e^{-\frac{3}{2}}}$
A1	$e^{x+\frac{7}{2}}$ oe
ALT	
M1	For $a = \frac{e^{x+2}}{e^{-\frac{3}{2}}}$
M1	Using $e^{a+b} = e^a \times e^b$ to simplify to $e^{x+2+\frac{3}{2}}$
A1	$e^{x+\frac{7}{2}}$ oe

Type MS · Finding n in a geometric series with logarithms

Answer 9 of 14 · Question 7 part d · 4 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 4 · find part d in this mark scheme extract

(c)	$S_\infty = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} = \dfrac{e^{x+\frac{7}{2}}}{\frac{e^{\frac{1}{2}}-1}{e^{\frac{1}{2}}}} = \dfrac{e^{x+4}}{e^{\frac{1}{2}}-1} \Rightarrow p = x+4$ **ALT 1** $S_\infty = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} \times \dfrac{e^{\frac{1}{2}}}{e^{\frac{1}{2}}} = \dfrac{e^{x+4}}{e^{\frac{1}{2}}-1} \Rightarrow p = x+4$ **ALT 2** $S_\infty = \dfrac{e^{x+\frac{7}{2}}}{1-e^{-\frac{1}{2}}} = \dfrac{e^{p}}{e^{\frac{1}{2}}-1} \Rightarrow e^{x+\frac{7}{2}}\left(e^{\frac{1}{2}}-1\right) = e^{p}\left(1-e^{-\frac{1}{2}}\right)$ $\Rightarrow e^{x+4} - e^{x+\frac{7}{2}} = e^{p} - e^{p-\frac{1}{2}} \Rightarrow p = x+4$	M1M1A1 [3] {M1}{M1} {A1} [3] {M1}{M1} {A1} [3]

(c)	
M1	Use of $S_\infty = \dfrac{a}{1-r}$
M1	$1-e^{-\frac{1}{2}} = \dfrac{e^{\frac{1}{2}}-1}{e^{\frac{1}{2}}}$
A1	$p = x+4$
ALT 1	
M1	Use of $S_\infty = \dfrac{a}{1-r}$
M1	Multiplying S_∞ by $\dfrac{e^{\frac{1}{2}}}{e^{\frac{1}{2}}}$
A1	$p = x+4$
ALT 2	
M1	Use of $S_\infty = \dfrac{a}{1-r}$
M1	For simplifying to $e^{x+4} - e^{x+\frac{7}{2}} = e^{p} - e^{p-\frac{1}{2}}$ oe
A1	$p = x+4$

Type MS · Finding n in a geometric series with logarithms

Answer 9 of 14 · Question 7 part d · 4 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 4 · find part d in this mark scheme extract

(d)	$e^{x+\frac{7}{2}} \times \left(e^{-\frac{1}{2}}\right)^{17} > 1.6 \Rightarrow e^x \times e^{\left(\frac{7}{2}-\frac{17}{2}\right)} > 1.6 \Rightarrow e^x > \frac{1.6}{e^{-5}}$	M1
	$\Rightarrow e^x > 237.46105....$	M1
	$\Rightarrow x > \ln(237.46105) \Rightarrow x > 5.4700...$	M1
	$\Rightarrow x = 6$	A1
		[4]
	ALT	
	$e^{x+\frac{7}{2}} \times \left(e^{-\frac{1}{2}}\right)^{17} > 1.6 \Rightarrow e^{x-5} > 1.6$	{M1}
	$\Rightarrow x - 5 > \ln(1.6)$	{M1}
	$\Rightarrow x > \ln(1.6) + 5 \Rightarrow x > 5.4700...$	{M1}
	$\Rightarrow x = 6$	{A1}
		[4]
		Total 13 marks

Type MS · Finding n in a geometric series with logarithms

Answer 9 of 14 · Question 7 part d · 4 marks

2020 JAN · Paper 02 (regional R) · extract 4 of 4 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding n in a geometric series with logarithms

Answer 10 of 14 · Question 7 part b · 2 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

7 (a)	$\frac{61}{6}=\frac{a\left(1-r^3\right)}{1-r}$ $\rightarrow$ **1**	M1
	$\frac{125}{6}=\frac{a}{1-r}$ $\rightarrow$ **2**	M1
(i)	$1\div 2\Rightarrow \frac{\frac{61}{6}}{\frac{125}{6}}=\frac{\frac{a\left(1-r^3\right)}{1-r}}{\frac{a}{1-r}}\Rightarrow \frac{61}{125}=1-r^3\Rightarrow r=\frac{4}{5}*$	dM1A1 cso
(ii)	$\frac{125}{6}=\frac{a}{1-\frac{4}{5}}\Rightarrow a=\frac{25}{6}$	M1A1
		[6]
	ALT	
	$a+ar+ar^2=a\ 1+r+r^2\ =\frac{61}{6}$ and $\frac{a}{1-r}=\frac{125}{6}$	[M1M1
	$\Rightarrow \frac{125}{6}\ 1-r\ \ 1+r+r^2\ =\frac{61}{6}\Rightarrow 125\ 1-r^3\ =61$	dM1
(i)	$\Rightarrow 1-r^3=\frac{61}{125}\Rightarrow r^3=\frac{64}{125}\Rightarrow r=\frac{4}{5}*$	A1 cso
(ii)	$\frac{125}{6}=\frac{a}{1-\frac{4}{5}}\Rightarrow a=\frac{25}{6}$	M1A1]
(b)	$19.8<\frac{\frac{25}{6}\left(1-0.8^n\right)}{1-0.8}\Rightarrow \frac{19.8\times 6\times 0.2}{25}<1-0.8^n$	M1
	$\frac{594}{625}<1-0.8^n\Rightarrow 0.8^n<\frac{31}{625}$	
	$n\lg(0.8)<\lg\left(\frac{31}{625}\right)*$	A1 cso
		[2]
(c)	$n>\frac{\lg\left(\frac{31}{625}\right)}{\lg(0.8)}\Rightarrow n>13.461...\Rightarrow n=14$	M1A1
		[2]
		Total 10 marks

Type MS · Finding n in a geometric series with logarithms

Answer 10 of 14 · Question 7 part b · 2 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

	M1	For both $\frac{61}{6}=\frac{a\left(1-r^3\right)}{1-r}$ and $\frac{125}{6}=\frac{a}{1-r}$ correct
	dM1	For a valid method to eliminate a from both equations by division or substitution and attempting to re-arrange to find a value for r^3. Eg., $\frac{125}{6}=\frac{a}{1-r}\Rightarrow a=\frac{125(1-r)}{6}\Rightarrow\frac{61}{6}=\frac{\left[\frac{125(1-r)}{6}\right]\left(1-r^3\right)}{1-r}\Rightarrow r^3=1-\frac{61}{125}$ This mark is dependent on **both** previous M marks.
	A1*	For $r=\frac{4}{5}$ This value is given, there must be no errors for the award of this mark
	ALT	
(i)	**M1**	For either $a+ar+ar^2=\left[a\ 1+r+r^2\ \right]=\frac{61}{6}$ or $\frac{125}{6}=\frac{a}{1-r}$ correct
	M1	For both $a+ar+ar^2=\left[a\ 1+r+r^2\ \right]=\frac{61}{6}$ and $\frac{125}{6}=\frac{a}{1-r}$ correct
	dM1	For a valid method to eliminate a from both equations by division or substitution and attempting to re-arrange to find a value for r^3. E.g. $\Rightarrow\frac{125}{6}\ 1-r\ \ 1+r+r^2\ =\frac{61}{6}\Rightarrow 125\ 1-r^3\ =61\Rightarrow r^3=1-\frac{61}{125}$ This mark is dependent on **both** previous M marks.
	A1	For $r=\frac{4}{5}$ This value is given, there must be no errors for the award of this mark
	M1	For substituting $\frac{4}{5}$ into either **their** $\frac{61}{6}=\frac{a\left(1-r^3\right)}{1-r}$ or **their** $\frac{125}{6}=\frac{a}{1-r}$ or **their** $a+ar+ar^2=\frac{61}{6}$ (where their expression for a sum has been seen earlier) to find a value of a
	A1	For $a=\frac{25}{6}$
(b)	**M1**	For using the **correct** formula for the sum of a geometric series with $r=0.8$ [oe] and their a, setting up an inequality (allow < or > for this mark) using the value of 19.8 and attempting to achieve the given result.
	A1*	For the correct inequality as shown with no errors as this is a given result. $n\lg(0.8)<\lg\left(\frac{31}{625}\right)$
(c)	**M1**	For solving the given inequality in n using logarithms. They must achieve a value for n for this mark. Allow use of <, >, or = for this mark.
	A1	For $n=14$

Type MS · Finding n in a geometric series with logarithms

Answer 10 of 14 · Question 7 part b · 2 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding n in a geometric series with logarithms

Answer 11 of 14 · Question 7 part f · 4 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part f in this mark scheme extract

7 (a)	$64\left(\frac{1}{4}\right)^3 - 64\left(\frac{1}{4}\right)^2 + 3 = 0$ so $(4x-1)$ is a factor	M1 A1 [2]
(b)	$16x^2 - 12x + "-3"$ $4x-1 \overline{)64x^3 - 64x^2 \{+0x\} + 3}$ $64x^3 - 16x^2$ $-48x^2$ or $64x^3 - 64x^2 + 3 = (4x-1)(16x^2 - 12x + C)$	M1
	$x = \frac{--12 \pm \sqrt{(-12)^2 - 4(16)(-3)}}{2\times 16}$	dM1
	$x = \frac{1}{4}$ oe	B1
	$x = \frac{3 \pm \sqrt{21}}{8}$ oe	A1 [4]
(c)	$ar^2 = 9$ or $\frac{a}{1-r} = 192$ oe	B1
	$(192 - 192r)r^2 = 9$ oe	M1
	$192r^2 - 192r^3 = 9 \Rightarrow 64r^3 - 64r^2 + 3 = 0$	A1*cso [3]
(d)	$r = \frac{1}{4}$	B1 [1]
(e)	$a\left(\frac{1}{4}\right)^2 = 9$ or $\frac{a}{1-\frac{1}{4}} = 192$ $\Rightarrow a = ...$	M1
	$a = 144$	A1*cso [2]
(f)	$\frac{144\left(1-0.25^n\right)}{1-0.25} > 191.9$	M1
	$1 - 0.25^n > \frac{1919}{1920}$	
	$0.25^n < 0.00052083...$	dM1
	$n > \frac{\log 0.00052083}{\log 0.25}$	M1
	$n > 5.45... \Rightarrow n = 6$	A1 [4]
		Total 16 marks

Type MS · Finding n in a geometric series with logarithms

Answer 11 of 14 · Question 7 part f · 4 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part f in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For $64\left(\pm\frac{1}{4}\right)^3-64\left(\pm\frac{1}{4}\right)^2+3$
	A1	For = 0 and a conclusion stated, no errors seen. Accept minimal conclusion.
(b)	M1	Attempts long division. Minimally acceptable attempt is the division and correct working as written in the scheme. Must get a 3TQ factor. If comparing coefficients, a correct equation/comparison must be written followed by an attempt to find *C*. Must get a 3TQ factor.
	dM1	For a fully correct method to solve their 3TQ to find *x*, correct solutions from a correct 3TQ implies this mark.
	B1	For correct root $\frac{1}{4}$
	A1	For correct roots $\frac{3\pm\sqrt{21}}{8}$
(c)	B1	For either correct equation.
	M1	For both correct equations and any correct step to eliminate a
	A1*cso	For a fully correct solution, no errors, with at least one correct intermediate line leading to $64r^3-64r^2+3=0$
(d)	B1 **(A1 on ePen)**	For the correct value of r
(e)	M1	For correct substitution of their value for r in either of their equations to find a
	A1	For $a = 144$
(f)	M1	**Allow working in equation or any inequality for all M marks** Correct use of sum to n terms formula with $a = 144$ or their a and their r to set up an equation or inequality
	dM1	Attempts to rearrange and obtains $\pm(\text{"their } r\text{"})^n \ldots p$
	M1	Correct use of logs and attempt to evaluate the log. (Can be implied by awrt 5.5 only)
	A1	$n = 6$ (must come from solving inequalities with no errors in processing the inequalities. If from solving equations only, award A0)

Type MS · Finding n in a geometric series with logarithms

Answer 11 of 14 · Question 7 part f · 4 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part f in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding n in a geometric series with logarithms

Answer 12 of 14 · Question 7 part c · 5 marks

2023 NOV · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

7(a)	$\left(\frac{ar^2}{a}=r^2=\right)\frac{2704}{625}$ oe	M1
	$(r=)\pm\frac{13}{25}$ oe	A1 [2]
(b)	$(S_\infty=)\frac{16}{1-"\frac{13}{25}"}=\frac{100}{3}$ oe	M1A1 [2]
(c)	$\frac{16\left(1-\left("\frac{13}{25}"\right)^n\right)}{1-"\frac{13}{25}"}>33\left(\Rightarrow 1-\left(\frac{13}{25}\right)^n>0.99\right)$ oe	M1
	$\Rightarrow\left("\frac{13}{25}"\right)^n<0.01$ oe	dM1
	$n\lg\left("\frac{13}{25}"\right)<\lg(0.01)$ oe	M1
	$\Rightarrow n>\frac{\lg(0.01)}{\lg\left("\frac{13}{25}"\right)}\Rightarrow n>7.04...\text{oe}\Rightarrow(n=)8$	dddM1A1 [5]
		Total 9 marks

Type MS · Finding n in a geometric series with logarithms

Answer 12 of 14 · Question 7 part c · 5 marks

2023 NOV · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For any correct unsimplified expression for r or r^2.
	A1	For $r = \pm\frac{13}{25}$ oe
(b)	M1	Uses the correct formula for the sum to infinity of a geometric series using their positive $r < 1$ and the given first term.
	A1	For $\frac{100}{3}$ oe . Allow 33.33……, 33.3^r, $33.\dot{3}$ etc. The question demand is for an exact answer so there must be some indication of recurrence. This could be min 1 dp with dots after ie 33.3….., but cannot be a terminating decimal.
(c)	M1	Uses the correct formula for the sum of a geometric series to set up an **inequality or equation** in terms of n, *ft* their **positive** $r < 1$, must be using $a = 16$ Allow < or > or =
	dM1	For simplifying (allow errors in simplification) their **inequality or equation** in n to the form $\left("\frac{13}{25}"\right)^n < d \quad d \neq 0$. Allow $"\frac{13}{25}"$ to be their positive $r < 1$. Allow < or > or =. Dependent on the 1st method mark.
	M1	Takes logarithms (any base) of their **exponential equation or inequality <u>correctly</u>** on both sides and correctly uses the power law to reach $n\lg(a) < \lg(b)$. or 'de-logs' their **exponential equation or inequality <u>correctly</u>** to get $\log_e d < e$ This isn't a dependent mark, but the candidate must be able to correctly take logs of both sides and use the power law or correctly de-log their equation/inequality. r can be any value carried through from part (a) for this mark. Allow < or > or =, the inequality doesn't need to have been reversed at this point.
	dddM1	For finding a value for n setting up the correct inequality (with their positive $r < 1$) from the beginning and where the inequality sign has been reversed at the appropriate point during their work. Dependent on all previous method marks. If candidates give a final answer of (n =) 8 – this mark can be implied even if the inequality sign is not correctly reversed.
	A1	For (n =) 8

Type MS · Finding n in a geometric series with logarithms

Answer 12 of 14 · Question 7 part c · 5 marks

2023 NOV · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding n in a geometric series with logarithms

Answer 13 of 14 · Question 7 part c · 5 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

7(a)	(i) $a = 9$	B1
	(ii) $d = 4$	B1 (2)
(b)	(i) $a = 4$	B1
	(ii) $r = 3$	B1 (2)
(c)	$A_{14} = \frac{14}{2}(2\times 9+13\times 4)$ or $\frac{14}{2}(9+61)$, $= 490$	M1, A1
	$"490"-6 = \frac{4(3^n-1)}{3-1}$	M1
	$3^n = 243$ $n = 5$	ddM1A1 (5) [9]

(a)	
B1	Correct value, no working or explanation needed
B1	Correct value, no working or explanation needed
(b)	
B1	Correct value, no working or explanation needed
B1	Correct value, no working or explanation needed
(c)	
M1	Use either formula for the sum of an arithmetic series with their a and d (if needed) and obtain a value for the sum of the first 14 terms
A1	Correct value for the sum
M1	Subtract 6 from their sum (explicitly or implicitly) and equate to the sum of the first n terms of the geometric series obtained using their a and r
ddM1	Solve their equation by a correct method. No method need be shown but must reach n = ... Depends on both M marks above
A1	Correct value for n obtained
ALT	For the last 3 marks:
M1	Subtract 6 from their sum and generate at least the first 5 terms of the geometric series.
ddM1	Sum their terms until at least “484” is reached
A1	Correct answer (5) obtained from correct work.

Type MS · Finding n in a geometric series with logarithms

Answer 13 of 14 · Question 7 part c · 5 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Finding n in a geometric series with logarithms

Answer 14 of 14 · Question 7 part c · 2 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

7 (a)	$\frac{61}{6}=\frac{a\left(1-r^3\right)}{1-r}$ $\rightarrow$ **1**	M1
	$\frac{125}{6}=\frac{a}{1-r}$ $\rightarrow$ **2**	M1
(i)	$1\div 2\Rightarrow \frac{\frac{61}{6}}{\frac{125}{6}}=\frac{\frac{a\left(1-r^3\right)}{1-r}}{\frac{a}{1-r}}\Rightarrow \frac{61}{125}=1-r^3\Rightarrow r=\frac{4}{5}*$	dM1A1 cso
(ii)	$\frac{125}{6}=\frac{a}{1-\frac{4}{5}}\Rightarrow a=\frac{25}{6}$	M1A1
		[6]
	ALT	
	$a+ar+ar^2=a\ 1+r+r^2\ =\frac{61}{6}$ and $\frac{a}{1-r}=\frac{125}{6}$	[M1M1
	$\Rightarrow \frac{125}{6}\ 1-r\ \ 1+r+r^2\ =\frac{61}{6}\Rightarrow 125\ 1-r^3\ =61$	dM1
(i)	$\Rightarrow 1-r^3=\frac{61}{125}\Rightarrow r^3=\frac{64}{125}\Rightarrow r=\frac{4}{5}*$	A1 cso
(ii)	$\frac{125}{6}=\frac{a}{1-\frac{4}{5}}\Rightarrow a=\frac{25}{6}$	M1A1]
(b)	$19.8<\frac{\frac{25}{6}\left(1-0.8^n\right)}{1-0.8}\Rightarrow \frac{19.8\times 6\times 0.2}{25}<1-0.8^n$	M1
	$\frac{594}{625}<1-0.8^n\Rightarrow 0.8^n<\frac{31}{625}$	
	$n\lg(0.8)<\lg\left(\frac{31}{625}\right)*$	A1 cso
		[2]
(c)	$n>\frac{\lg\left(\frac{31}{625}\right)}{\lg(0.8)}\Rightarrow n>13.461...\Rightarrow n=14$	M1A1
		[2]
		Total 10 marks

Type MS · Finding n in a geometric series with logarithms

Answer 14 of 14 · Question 7 part c · 2 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

	M1	For both $\frac{61}{6}=\frac{a\left(1-r^3\right)}{1-r}$ and $\frac{125}{6}=\frac{a}{1-r}$ correct
	dM1	For a valid method to eliminate a from both equations by division or substitution and attempting to re-arrange to find a value for r^3. Eg., $\frac{125}{6}=\frac{a}{1-r}\Rightarrow a=\frac{125(1-r)}{6}\Rightarrow\frac{61}{6}=\frac{\left[\frac{125(1-r)}{6}\right]\left(1-r^3\right)}{1-r}\Rightarrow r^3=1-\frac{61}{125}$ This mark is dependent on **both** previous M marks.
	A1*	For $r=\frac{4}{5}$ This value is given, there must be no errors for the award of this mark
	ALT	
(i)	**M1**	For either $a+ar+ar^2=\left[a\ 1+r+r^2\ \right]=\frac{61}{6}$ or $\frac{125}{6}=\frac{a}{1-r}$ correct
	M1	For both $a+ar+ar^2=\left[a\ 1+r+r^2\ \right]=\frac{61}{6}$ and $\frac{125}{6}=\frac{a}{1-r}$ correct
	dM1	For a valid method to eliminate a from both equations by division or substitution and attempting to re-arrange to find a value for r^3. E.g. $\Rightarrow\frac{125}{6}\ 1-r\ \ 1+r+r^2\ =\frac{61}{6}\Rightarrow 125\ 1-r^3\ =61\Rightarrow r^3=1-\frac{61}{125}$ This mark is dependent on **both** previous M marks.
	A1	For $r=\frac{4}{5}$ This value is given, there must be no errors for the award of this mark
	M1	For substituting $\frac{4}{5}$ into either **their** $\frac{61}{6}=\frac{a\left(1-r^3\right)}{1-r}$ or **their** $\frac{125}{6}=\frac{a}{1-r}$ or **their** $a+ar+ar^2=\frac{61}{6}$ (where their expression for a sum has been seen earlier) to find a value of a
	A1	For $a=\frac{25}{6}$
(b)	**M1**	For using the **correct** formula for the sum of a geometric series with $r=0.8$ [oe] and their a, setting up an inequality (allow < or > for this mark) using the value of 19.8 and attempting to achieve the given result.
	A1*	For the correct inequality as shown with no errors as this is a given result. $n\lg(0.8)<\lg\left(\frac{31}{625}\right)$
(c)	**M1**	For solving the given inequality in n using logarithms. They must achieve a value for n for this mark. Allow use of <, >, or = for this mark.
	A1	For $n=14$

Type MS · Finding n in a geometric series with logarithms

Answer 14 of 14 · Question 7 part c · 2 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks